

MUSCAT INTL, Oman

WMO: 412560

Lat: 23.6086N

Lon: 58.2614E

Elev: 28

StdP: 14.68

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.7	61.5	31.9	26.3	89.8	36.0	31.0	86.4	20.0	76.3	17.7	74.9	3.6	200	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.6	108.6	72.1	106.0	72.8	103.9	73.2	87.0	93.1	86.1	92.4	85.3	91.8	10.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
85.8	189.2	91.1	84.5	181.2	90.5	83.9	177.9	90.3	52.1	92.5	51.1	92.9	50.1	91.4	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.8	14.8	DB	55.4	114.9	1.6	2.6	54.2	116.7	53.2	118.2	52.3	119.6	51.2	121.5	(4)
(5)			WB	46.9	88.9	1.8	0.9	45.6	89.5	44.5	90.0	43.5	90.5	42.3	91.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	69.6	71.9	76.9	85.2	93.7	94.8	92.2	89.5	88.1	85.5	78.0	72.1
	DBStd	9.54	2.81	3.77	4.38	5.10	5.01	4.72	4.40	4.33	3.29	3.32	3.29	3.03
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	3	2	1	0	0	0	0	0	0	0	0	0	0
	CDD50	12106	608	612	835	1057	1355	1343	1307	1223	1142	1099	839	686
	CDD65	6635	145	193	370	607	890	893	842	758	692	634	389	222
	CDH74	89040	488	972	3115	8160	14573	14685	13099	11082	9656	8421	3645	1144
	CDH80	52259	25	156	960	4401	10133	10371	8637	6634	5389	4351	1090	112
Wind	WSAvg	6.1	5.7	6.2	6.4	6.3	7.0	6.8	6.7	6.3	5.7	5.5	5.4	5.3
Precipitation	PrecAvg	3.4	0.7	0.7	0.4	0.3	0.1	0.0	0.1	0.0	0.0	0.0	0.3	0.4
	PrecMax	9.4	4.0	3.4	5.7	2.6	2.7	0.4	2.7	1.1	0.1	0.5	3.0	4.4
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.6	0.9	0.9	0.9	0.6	0.4	0.0	0.4	0.2	0.0	0.1	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.6	89.3	95.3	104.5	112.6	112.8	109.5	107.8	103.9	101.5	91.9	85.9
		MCWB	65.5	64.9	66.5	68.8	70.7	72.3	74.0	74.0	72.6	69.9	69.5	66.3
	2%	DB	80.2	84.3	90.5	100.6	107.9	109.0	105.8	103.8	100.2	98.2	89.0	82.6
		MCWB	66.6	66.9	66.2	68.6	70.5	72.2	75.3	75.9	73.9	70.8	71.2	69.3
	5%	DB	78.5	80.9	87.7	97.2	105.6	106.0	102.7	100.4	97.0	95.2	86.9	80.8
		MCWB	66.8	67.5	67.0	69.1	71.1	73.2	75.8	76.6	75.9	72.3	73.0	69.5
	10%	DB	76.8	79.0	85.1	94.9	102.6	103.8	100.0	96.7	94.4	93.0	85.7	79.2
		MCWB	66.5	67.6	67.9	69.2	71.4	73.9	77.8	79.4	77.8	73.8	73.0	68.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	74.0	77.6	80.9	85.9	88.0	88.2	87.1	85.8	83.6	78.3	74.3
		MCDB	77.5	79.2	82.7	88.2	92.3	94.0	93.9	93.2	91.3	89.8	85.8	80.4
	2%	WB	70.1	72.4	75.6	79.0	84.2	86.9	87.2	85.8	84.5	81.8	77.0	72.6
		MCDB	76.3	78.0	81.2	86.3	91.5	93.2	93.0	92.0	90.3	88.5	84.9	79.2
	5%	WB	69.1	71.3	74.3	77.6	82.9	85.9	86.2	84.8	83.6	80.7	75.6	71.4
		MCDB	75.6	77.3	80.1	85.5	90.5	92.6	92.3	91.0	89.3	87.9	84.0	78.3
	10%	WB	67.5	70.0	72.8	76.1	81.2	85.0	85.4	83.9	82.7	79.7	74.4	70.0
		MCDB	74.7	75.9	79.3	85.0	90.3	92.2	91.8	90.1	88.3	87.3	82.9	77.3
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.0	14.4	16.3	16.7	14.6	11.9	11.8	13.4	15.8	14.6	14.1
		MCDBR	15.9	16.7	18.5	20.3	19.9	19.0	17.3	17.5	18.1	19.2	16.2	15.7
		MCWBR	9.9	10.7	12.6	13.1	13.1	14.8	13.1	11.9	12.8	14.6	11.8	9.8
	5% WB	MCDBR	14.5	13.9	13.7	15.6	15.0	13.7	12.1	12.5	13.1	14.0	14.2	14.1
	MCWBR	9.8	9.6	10.6	12.5	13.4	12.6	9.7	8.8	9.6	11.6	10.9	9.3	
Clear-Sky Solar Irradiance	taub	0.423	0.468	0.529	0.578	0.602	0.674	0.746	0.686	0.582	0.482	0.443	0.424	
	taud	2.229	2.073	1.900	1.817	1.768	1.658	1.571	1.664	1.833	2.044	2.182	2.218	
	Ebn at Noon	263	258	248	239	232	214	199	211	231	250	254	257	
	Edh at Noon	40	50	62	69	72	80	87	79	66	51	42	39	
All-Sky Solar Radiation	RadAvg	1414	1669	1941	2159	2362	2312	2243	2101	1979	1791	1487	1338	
	RadStd	69	76	102	114	53	78	88	56	39	48	44	68	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+0.41	N/A	N/A	N/A	+0.70	+0.81	N/A	N/A	+173
(47) Regional (12 neighbors)	DBAvg	+0.76	+1.22	N/A	N/A	+0.50	+0.54	N/A	N/A	+176

Nomenclature: See separate page